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classmate
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Maths

1. Monomial

2. $(x+1)(x^2 - x - x^4 + 1)$

$\Rightarrow x^3 - x^2 - 5x + x + x^2 - x - x^4 + 1$

$\Rightarrow$ degree of polynomial is 5.

3 $f(x) = a^2x^2 - 3ax + 3x - 1 = 0$

$\Rightarrow a^2(1)^2 - 3a(1) + 3(1) - 1 = 0$

$\Rightarrow a^2 - 3a + 3 - 1 = 0$

$\Rightarrow a^2 - 3a + 2 = 0$

$\Rightarrow a^2 - 2a \div a + 2 = 0$

$\Rightarrow a(a-2) - 1(a-2)$

$\Rightarrow (a-1)(a-2)$

$\Rightarrow a = 1, \quad a = 2.$

5. $f(x) = 2x^2 - 3x + k.$

here, let the zero be $\frac{1}{\alpha}$.

then $a = 2$
$b = -3$
$c = k.$

So $\alpha \times \frac{1}{\alpha} \Rightarrow \frac{c}{a}$

$1 \Rightarrow \frac{k}{2}$

the $k = 2.$

6. If α and β are the zeros of the polynomial then $\alpha + \beta = 5$ 8

i.e $\frac{-b}{a} = 5$

and $\alpha\beta = -14$

$\frac{c}{a} = -14.$

$(x-\alpha)(x-\beta)$ is a quadratic polynomial

$x\beta \quad x\alpha.$

$\Rightarrow x^2 - \cancel{\alpha x} - \cancel{\beta x} + \alpha\beta$

$\Rightarrow x^2 - (\alpha+\beta)x + \alpha\beta.$ 9

$\Rightarrow x^2 - 5x + (-14)$

$\Rightarrow \boxed{x^2 - 5x - 14.}$

7. a $p(x) = x^2 + 5x + 6$

$\Rightarrow x^2 + 3x + 2x + 6$

$\Rightarrow x(x+3) + 2(x+3)$

$\Rightarrow (x+2)(x+3)$

$\Rightarrow x = -2, \quad x = -3$

Now Verification:

L.H.S Sum of zeros $= -2 + (-3) \Rightarrow -2 - 3 = -5$

R.H.S, $\alpha + \beta = \frac{-b}{a} \Rightarrow \frac{-5}{1} \Rightarrow -5$

Hence Proved

L.H.S Product of zeros $= -2 \times -3 \Rightarrow 6$

R.H.S $\frac{\alpha}{\beta} = \frac{c}{a} = \frac{6}{1} = 6$

Hence Proved

8 Given zeros are 3 , 5 and -2

$\Rightarrow (x-3)(x-5)(x+2)$

$\Rightarrow$ On simplifying we get ,

$(x-3)(x-5)(x+2)$

$\Rightarrow (x^2-8x+15)(x+2)$

$\Rightarrow x^3-6x^2-x+30$

9. $x+\frac{1}{x} = 3$ (on cubing both sides)

$\cancel{x^3}\left[x+\frac{1}{x}\right]^3 = (3)^3$

$\Rightarrow (x)^3+\left[\frac{1}{x}\right]^3+3(\cancel{x})\left(\frac{1}{\cancel{x}}\right)\left(x+\frac{1}{x}\right) = 27$ $\left[\because (a+b)^3 \Rightarrow a^3+b^3+3ab(a+b)\right.$

$\Rightarrow x^3+\frac{1}{x^3}+9 = 27$ $\therefore \left[\text{Given, } x+\frac{1}{x} = 3\right.$

$\Rightarrow x^3+\frac{1}{x} = 27-9$

$\Rightarrow \boxed{x^3+\frac{1}{x} = 18.}$

10 If (α and β) are the zeros of the polynomial $15x^2 - 5x + 6$ then,

$$(\alpha + \beta) = \frac{-(b)}{a} \Rightarrow \frac{-(-5)}{15} = \frac{5}{15} = \frac{1}{3}$$

$$(\alpha\beta) = \frac{c}{a} \Rightarrow \frac{6}{15} = \frac{2}{5}$$

now, $\left(1 + \frac{1}{\alpha}\right)\left(1 + \frac{1}{\beta}\right) \Rightarrow \left(\frac{\alpha + 1}{\alpha}\right)\left(\frac{\beta + 1}{\beta}\right)$

$$\Rightarrow \frac{\alpha\beta + \alpha + \beta + 1}{\alpha\beta}$$

$$\Rightarrow \frac{\frac{2}{5} + \frac{1}{3} + 1}{\frac{2}{5}}$$

$$\Rightarrow \frac{\frac{6 + 5 + 15}{15}}{\frac{2}{5}} \Rightarrow \frac{\frac{26}{15}}{\frac{2}{5}}$$

$$\Rightarrow \frac{\cancel{26}^{13}}{\cancel{15}_{3}} \times \frac{\cancel{5}^{1}}{\cancel{2}_{1}}$$

$$\Rightarrow \boxed{\frac{13}{3}}$$